

Sameer Asif

AI / ML Engineer | LLMs · Computer Vision · Generative AI · Backend

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SUMMARY

AI/ML Engineer and final-year BS Artificial Intelligence student at FAST-NUCES, building production ML systems at Infigo Solutions. Experienced across the full ML lifecycle: data pipelines, PyTorch model training, and deployment via FastAPI and vector databases. Expertise in LLMs, RAG, computer vision, generative models (GANs, U-Nets), and speech processing, with a track record of shipping research prototypes to real-time production systems.

TECHNICAL SKILLS

- Languages:** Python, SQL, C/C++, JavaScript, TypeScript, Bash
- ML Frameworks:** PyTorch, Hugging Face Transformers, CNNs, GANs, U-Net, EfficientNet, multi-task learning, ONNX
- LLMs & NLP:** RAG, pgvector, embeddings, semantic search, prompt engineering, LoRA/SFT, language modeling
- CV & Speech:** OpenCV, timm, object detection, facial expression recognition, faster-whisper, Spark-TTS, librosa
- Backend & Data:** FastAPI, REST APIs, WebSockets, async Python, PostgreSQL/pgvector, SQL Server, SQLAlchemy, pandas, NumPy
- Frontend & Tools:** React, Vite, Git, Docker, Linux/WSL, Vercel, CUDA

EXPERIENCE

Junior AI Engineer

Infigo Solutions, Islamabad | July 2025 – Present

- Lead developer of a multi-channel AI customer support assistant (website, WhatsApp, Slack) using FastAPI and PostgreSQL (pgvector), live at [infigosolutions.com](#).
- Designed a RAG pipeline: document chunking, pgvector embeddings, cosine top-k search with full-text fallback, and ERP context (customers, orders, invoices) injected into model prompts.
- Added cost and safety controls around a Groq-hosted Llama 3.1 8B model: answer caching, token-budget tracking, retrieval limits, and escalation rules.
- Built an embeddable React chat widget and public chat API; authored database migrations, seed data, and smoke tests.

AI Engineering Intern

Infigo Solutions, Islamabad | June 2025 – July 2025

- Built an async web-scraping pipeline (Clutch, Wellfound) loading company and contact data into SQL Server for sales lead generation; added anti-bot handling and resumable progress tracking.
- Developed the initial support chatbot prototype with deployment and authentication flow, which led to a full-time offer.

PROJECTS

Adaptive Psychological Horror Game — Final Year Project

PyTorch, ONNX, Hugging Face, Unity · [GitHub](#)

- Trained facial expression recognition across 7 datasets (DFEW, FERV39k, FER2013, RAF-DB, CK+, Aff-Wild2, RAVDESS), fusing video, audio, and heart-rate signals into a real-time stress classifier.
- Built a stress-conditioned Fusion Generator (~500K params, sub-ms inference), WaveGAN (~2M params) for procedural audio, and a U-Net (~3M params, 256x256) for texture corruption; exported all to ONNX for Unity.

Voice-Cloning Conversational Agent

FastAPI, PyTorch, faster-whisper, Spark-TTS, React · [GitHub](#)

- Real-time voice assistant: faster-whisper STT on CUDA + Llama 3.1 8B (Groq) + Spark-TTS cloned-voice synthesis within 16 GB VRAM. WebSocket audio streaming backend and React/Vite frontend.

Hybrid Wildlife Detection

PyTorch, timm, albumentations, OpenCV · [GitHub](#)

- 4.5M-parameter detector with EfficientNet-B0 backbone, multi-scale feature fusion, spatial-attention gating, and three task heads (21-class, bbox regression, objectness). Trained on Missouri Camera Traps dataset.

Transformer & RAG from Scratch

PyTorch, NumPy · [GitHub](#)

- Multi-task Transformer encoder achieving 82% sentiment accuracy on Amazon reviews. Decoder with top-3 cosine RAG context; analyzed retrieval vs. no-retrieval perplexity trade-off.

Other Projects

- SIP Restaurant Ordering App**: React/Vite QR scan-to-order dining app with menu browsing, cart, and checkout, deployed on Vercel.
- BBC Urdu Text Generator**: from-scratch n-gram language models with a right-to-left Streamlit interface.
- Sentinel Binance Trading Bot**: sentiment-driven adaptive trading strategy that learns from past trades.

EDUCATION

BS Artificial Intelligence

FAST-NUCES, Islamabad | 2022 – 2026 (Expected)

Relevant coursework: Deep Learning, NLP, Computer Vision, Generative AI, Databases, Software Engineering, Data Structures & Algorithms, Probability & Statistics.

Cambridge A Levels & O Levels

Beaconhouse School System | 2017 – 2022